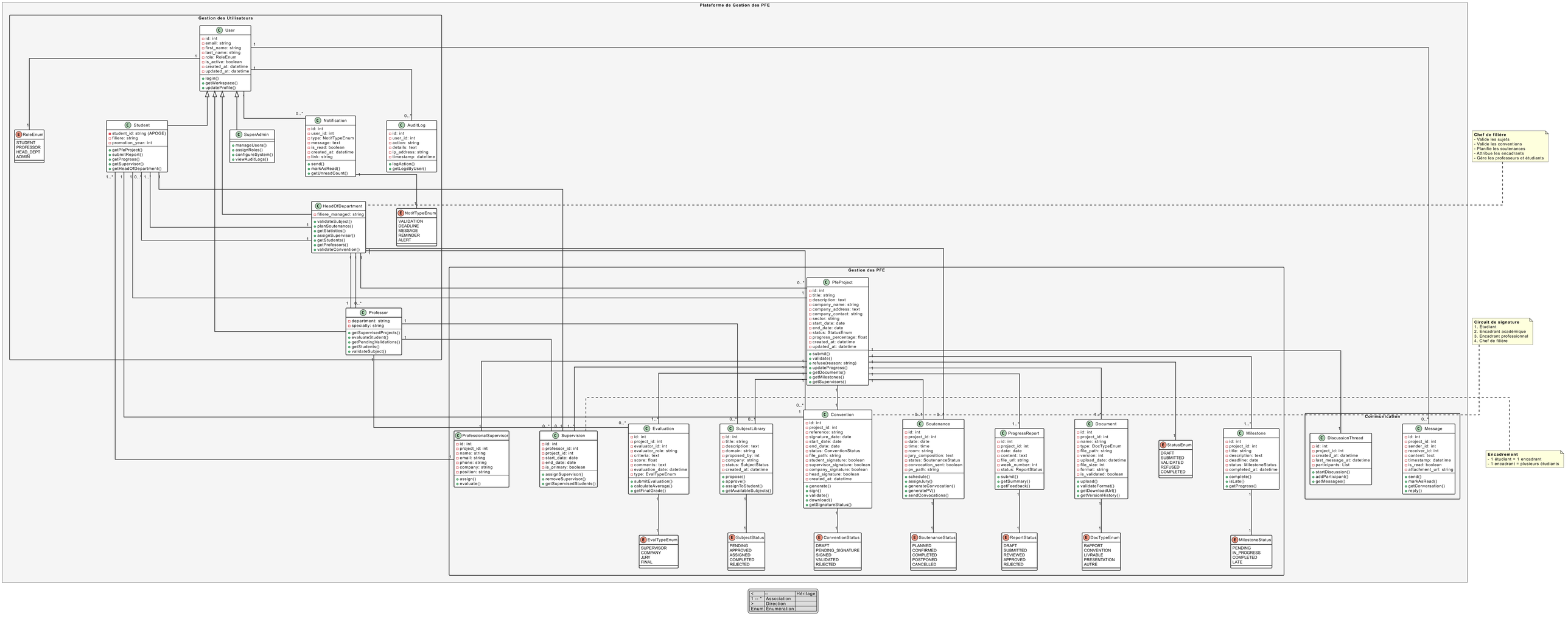
Plateforme de Gestion des PFE
Gestion des Utilisateurs
User
id: int
email: string
first_name: string
last_name: string
role: RoleEnum
is_active: boolean
created_at: datetime
updated_at: datetime
login()
getWorkspace()
updateProfile()
RoleEnum
STUDENT
PROFESSOR
HEAD_DEPT
ADMIN
Student
student_id: string (APOGE)
filiere: string
promotion_year: int
getPfeProject()
submitReport()
getProgress()
getSupervisor()
getHeadOfDepartment()
SuperAdmin
manageUsers()
assignRoles()
configureSystem()
viewAuditLogs()
Notification
id: int
user_id: int
type: NotifTypeEnum
message: text
is_read: boolean
created_at: datetime
link: string
send()
markAsRead()
getUnreadCount()
AuditLog
id: int
user_id: int
action: string
details: text
ip_address: string
timestamp: datetime
logAction()
getLogsByUser()
HeadOfDepartment
filiere_managed: string
validateSubject()
planSoutenance()
getStatistics()
assignSupervisor()
getStudents()
getProfessors()
validateConvention()
NotifTypeEnum
VALIDATION
DEADLINE
MESSAGE
REMINDER
ALERT
Professor
department: string
specialty: string
getSupervisedProjects()
evaluateStudent()
getPendingValidations()
getStudents()
validateSubject()
Chef de filière
- Valide les sujets
- Valide les conventions
- Planifie les soutenances
- Attribue les encadrants
- Gère les professeurs et étudiants
Gestion des PFE
PfeProject
id: int
title: string
description: text
company_name: string
company_address: text
company_contact: string
sector: string
start_date: date
end_date: date
status: StatusEnum
progress_percentage: float
created_at: datetime
updated_at: datetime
submit()
validate()
refuse(reason: string)
updateProgress()
getDocuments()
getMilestones()
getSupervisors()
Circuit de signature
1. Étudiant
2. Encadrant académique
3. Encadrant professionnel
4. Chef de filière
ProfessionalSupervisor
id: int
project_id: int
name: string
email: string
phone: string
company: string
position: string
assign()
evaluate()
Supervision
id: int
professor_id: int
project_id: int
start_date: date
end_date: date
is_primary: boolean
assignSupervisor()
removeSupervisor()
getSupervisedStudents()
Evaluation
id: int
project_id: int
evaluator_id: int
evaluator_role: string
criteria: text
score: float
comments: text
evaluation_date: datetime
type: EvalTypeEnum
submitEvaluation()
calculateAverage()
getFinalGrade()
EvalTypeEnum
SUPERVISOR
COMPANY
JURY
FINAL
SubjectLibrary
id: int
title: string
description: text
domain: string
proposed_by: int
company: string
status: SubjectStatus
created_at: datetime
propose()
approve()
assignToStudent()
getAvailableSubjects()
SubjectStatus
PENDING
APPROVED
ASSIGNED
COMPLETED
REJECTED
Convention
id: int
project_id: int
reference: string
signature_date: date
start_date: date
end_date: date
status: ConventionStatus
file_path: string
student_signature: boolean
supervisor_signature: boolean
company_signature: boolean
head_signature: boolean
created_at: datetime
generate()
sign()
validate()
download()
getSignatureStatus()
ConventionStatus
DRAFT
PENDING_SIGNATURE
SIGNED
VALIDATED
REJECTED
Soutenance
id: int
project_id: int
date: date
time: time
room: string
jury_composition: text
status: SoutenanceStatus
convocation_sent: boolean
pv_path: string
schedule()
assignJury()
generateConvocation()
generatePV()
sendConvocations()
SoutenanceStatus
PLANNED
CONFIRMED
COMPLETED
POSTPONED
CANCELLED
Encadrement
- 1 étudiant = 1 encadrant
- 1 encadrant = plusieurs étudiants
ProgressReport
id: int
project_id: int
date: date
content: text
file_url: string
week_number: int
status: ReportStatus
submit()
getSummary()
getFeedback()
ReportStatus
DRAFT
SUBMITTED
REVIEWED
APPROVED
REJECTED
Document
id: int
project_id: int
name: string
type: DocTypeEnum
file_path: string
version: int
upload_date: datetime
file_size: int
format: string
is_validated: boolean
upload()
validateFormat()
getDownloadUrl()
getVersionHistory()
DocTypeEnum
RAPPORT
CONVENTION
LIVRABLE
PRESENTATION
AUTRE
StatusEnum
DRAFT
SUBMITTED
VALIDATED
REFUSED
COMPLETED
Milestone
id: int
project_id: int
title: string
description: text
deadline: date
status: MilestoneStatus
completed_at: datetime
complete()
isLate()
getProgress()
MilestoneStatus
PENDING
IN_PROGRESS
COMPLETED
LATE
Communication
DiscussionThread
id: int
project_id: int
created_at: datetime
last_message_at: datetime
participants: List
startDiscussion()
addParticipant()
getMessages()
Message
id: int
project_id: int
sender_id: int
receiver_id: int
content: text
timestamp: datetime
is_read: boolean
attachment_url: string
send()
markAsRead()
getConversation()
reply()
< -- Héritage
1 -- * Association
> Direction
Enum Enumération